Roots of Polynomials: Using Substitutions

Questions 1–8

Question 1. The quadratic equation

$$3x^2 - 4x - 6 = 0$$

has roots α and β
In each case, without solving the equation find a quadratic equation with integer coefficients with the given roots:

a) $2\alpha, 2\beta$
b) $1 - \alpha, 1 - \beta$
c) $2\alpha + 3, 2\beta + 3$
d) $\frac{3}{\alpha}, \frac{3}{\beta}$

Question 2. The cubic equation

$$3z^3 - 3z^2 + 8z + 5 = 0$$

has roots α, β and γ
In each case, without solving the equation, find a cubic equation with integer coefficients with the given roots:

a) $2\alpha, 2\beta, 2\gamma$
b) $3 - \alpha, 3 - \beta, 3 - \gamma$
c) $2\alpha - 1, 2\beta - 1, 2\gamma - 1$
d) $\frac{\alpha}{3} + 1, \frac{\beta}{3} + 1, \frac{\gamma}{3} + 1$
e) $\frac{1}{2\alpha}, \frac{1}{2\beta}, \frac{1}{2\gamma}$

Question 3. The quartic equation

$$2z^4 - 2z^3 + z^2 + 4z + 7 = 0$$

has roots α, β, γ and δ
In each case, without solving the equation, find a quartic equation with integer coefficients with the given roots:

a) $2\alpha, 2\beta, 2\gamma, 2\delta$
b) $4-\alpha, 4-\beta, 4-\gamma, 4-\delta$
c) $2\alpha+3, 2\beta+3, 2\gamma+3, 2\delta+3$
d) $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}, \frac{1}{\delta}$

Question 4. The quadratic equation

$$2z^2 + 5z + k = 0$$

where k is a constant has roots α and β

a) Find an equation with roots $\frac{\alpha}{2} - 1$ and $\frac{\beta}{2} - 1$ giving your answer in terms of k
b) Given that $k = \frac{41}{8}$, use your answer to part a) to find the roots of $2z^2 + 5z + k = 0$

Question 5. The curve of

$$f(x) = x^3 + 2x^2 - 10x + k$$

where k is a positive integer has roots α, β and γ. The two stationary points of the curve are when $x = a$ and $x = b$

a) Find an expression for $g(x) = 0$ which is a cubic equation with roots $\alpha - 2$, $\beta - 2$ and $\gamma - 2$

b) State the x-coordinates of the stationary points of $g(x)$ in terms of a and b

Question 6. The cubic equation

$$2z^3 - z^2 + 3z + k = 0$$

where k is an integer has roots α, β and γ

a) Find an equation with roots $\frac{\alpha}{2} + 1$, $\frac{\beta}{2} + 1$ and $\frac{\gamma}{2} + 1$

When $k = 2$ it is known that the real solution to the equation you found in part a) is $\frac{3}{4}$

b) Find the real solution to $2z^3 - z^2 + 3z + k = 0$ for the given value of k

Question 7. The quadratic equation

$$x^2 + 5x - 7 = 0$$

has roots α and β

a) State the value of $\alpha\beta$

b) Hence, by making the substitution $x = \frac{y}{\alpha\beta}$ find an equation that has roots $\alpha^2\beta$ and $\alpha\beta^2$

Question 8. The roots of the equation

$$x^3 + kx^2 + (2k - 5)x + 10 = 0$$

are α, β and γ

Find an equation with roots $\alpha + 3$, $\beta + 3$ and $\gamma + 3$ giving your answer in the form

$$x^3 + p(k)x^2 + q(k)x + r(k) = 0$$